

Cognitive Modelling of Biological Agents

Project Proposal

Adam Krupinski & Sinem Altun



Agenda

01 Introduction: Problem Outline

02 Environment & Agent Design

03 Proposed Solutions

04 Evaluation: RL Metrics vs. Human Benchmarks

05 Work Plan

06 References

01 Introduction: Problem Outline

Why Study Human Behavior and Cognition?

- Humans constantly make decisions under uncertainty and risk
- Understanding these mechanisms helps explain:
 - learning
 - fear
 - exploration
 - and decision-making under stress

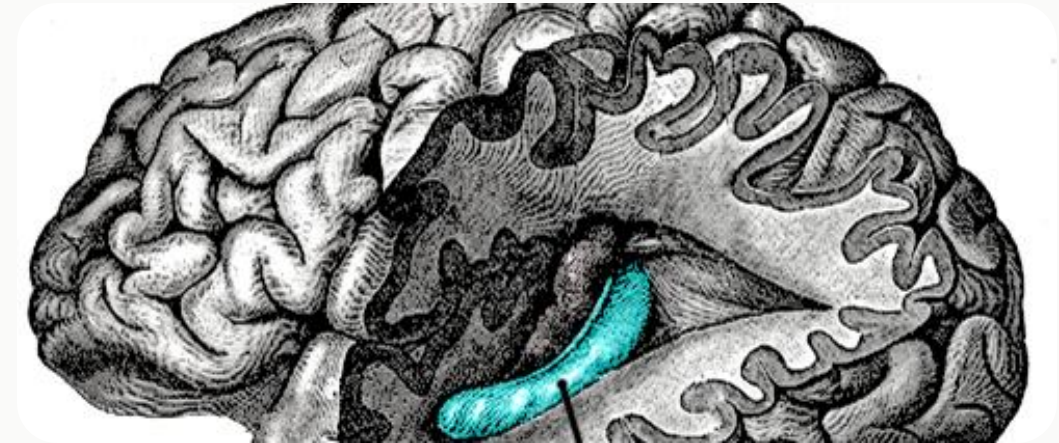
- We essentially want to gain a better understanding of our own capabilities
 - How does the brain structure itself?
 - How can mental capabilities arise?
 - How do we make sense of the world?
- Computational models allow us to test hypotheses systematically and efficiently

Our goal is not to create a perfect game-playing agent, but a biologically plausible model whose behavior can be compared to human decision-making.

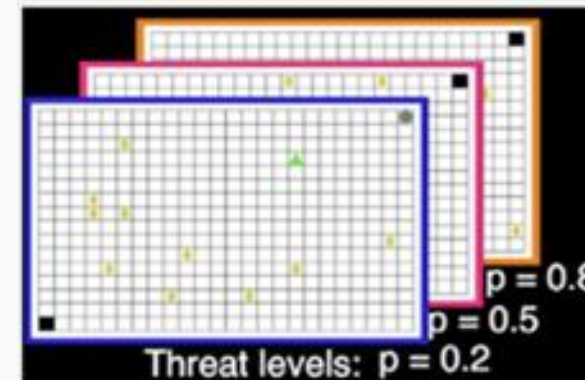
Approach-Avoidance Conflict

Decision making often involves an **approach-avoidance** conflict, which is mediated by the hippocampus.

Bach et al. investigated this using a 2D spatial game where participants collect tokens (**approach**) while risking a sleeping predator waking up (**avoidance**) across varying threat probabilities.



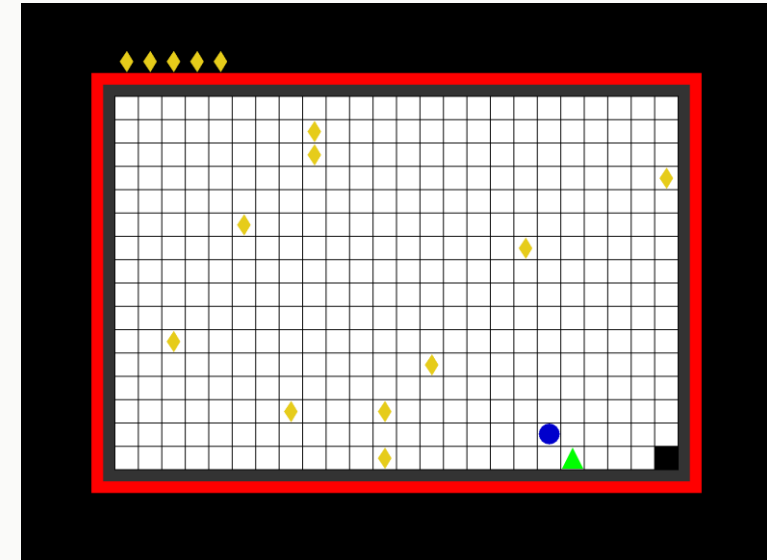
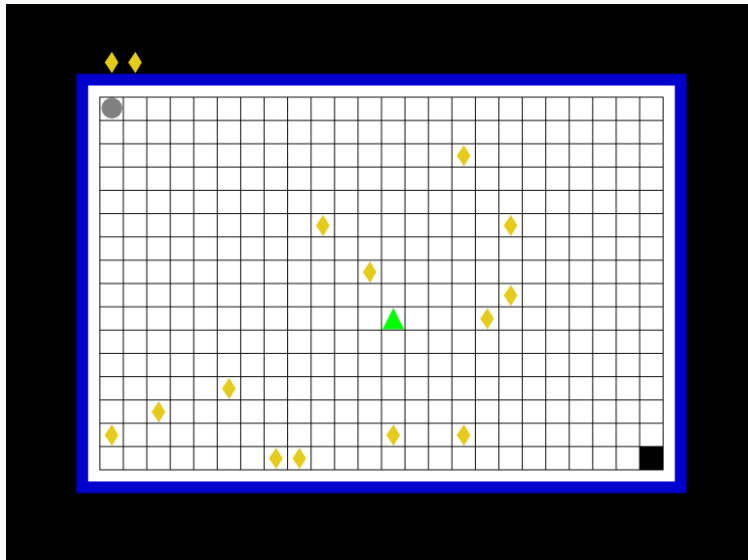
(1)



(2)

Overview – Study and Game Setting

Instructions given to study participants: Now you will play a computer game – think PacMan. On each round of the game you will walk over the grid for up to 20 seconds and collect tokens. It's important that you collect as many tokens as possible on each of the 240 rounds!



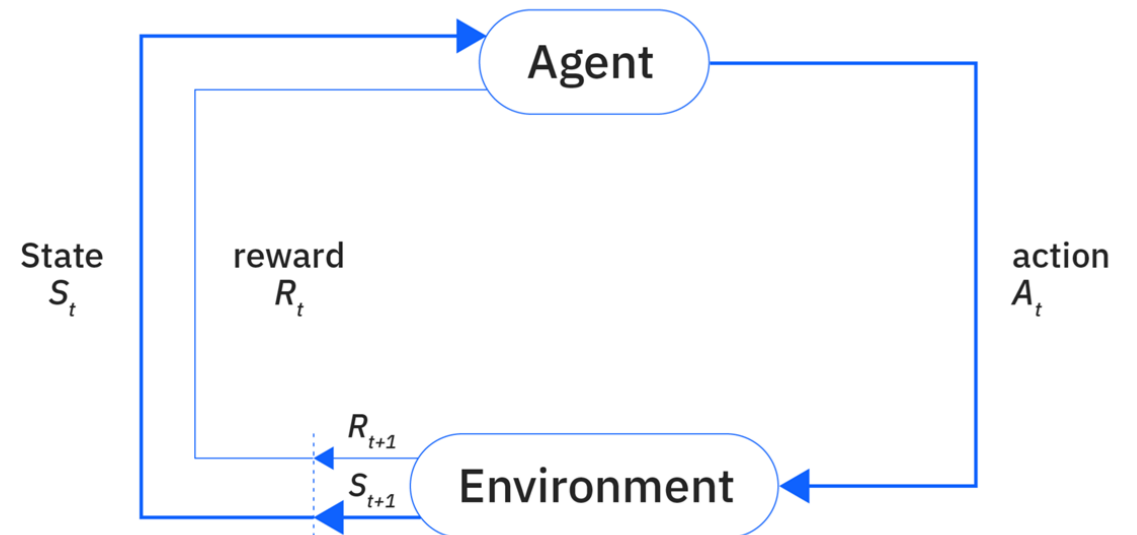
02 Environment & Agent Design

What is Reinforcement Learning (RL)?

Reinforcement Learning provides a computational framework for

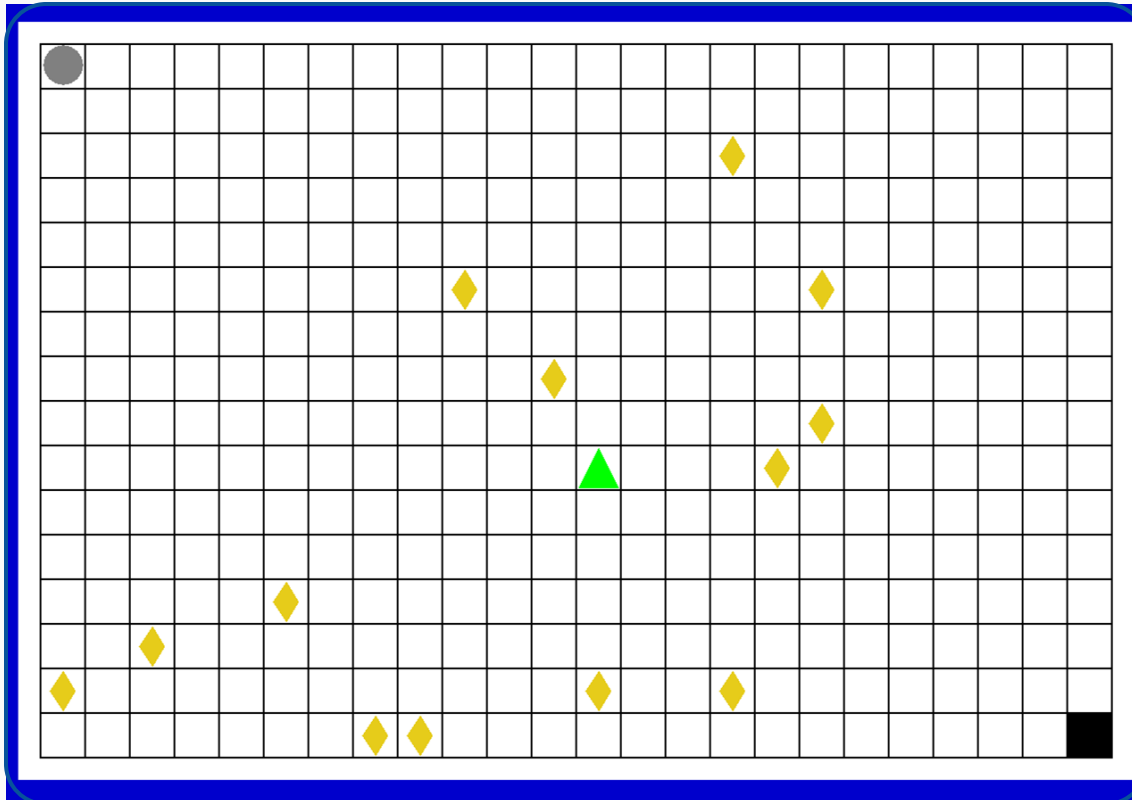
- learning from reward and punishment
- adapting behavior over time
- and modeling goal-directed decision making

Our Objective: Build Reinforcement Learning (RL) agents within this environment to simulate biological decision-making and compare model behavior against human performance data.



(3)

Environment Design



(4)

Spatial Grid (24x16)

Features: Agent (green triangle), Reward Tokens (yellow rhombi), Predator (grey circle), and Safe Zone (black rect).

Phase-Based Dynamics

Starts with a **Foraging phase** for collection. Predator wake-up triggers a **Chase phase** where the agent must reach the safe zone.

Threat Level Variables

Defined as low, medium, or high based on the probability of the predator waking up.

Technical Implementation

Built in Python with discrete action spaces (Up, Down, Left, Right) and balanced reward functions.

Agent Design – Basics

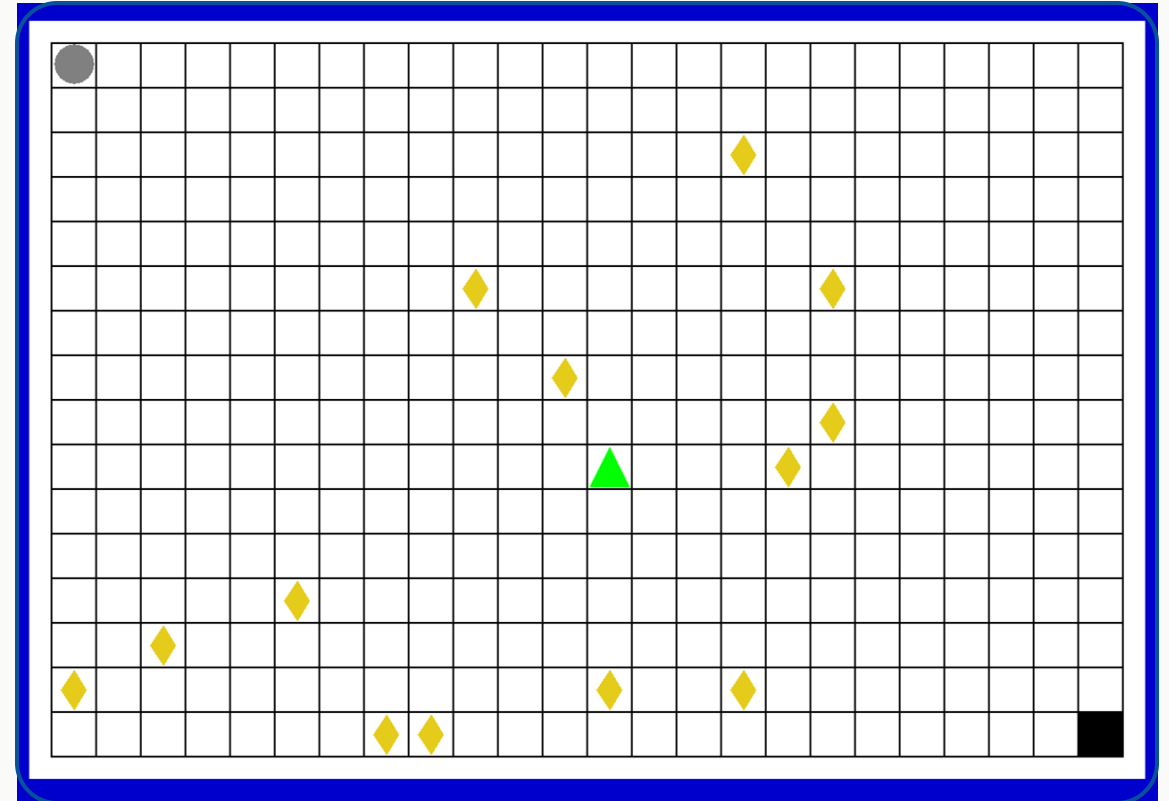
The agent must balance

- maximizing token collection
- avoiding predator capture
- and deciding when to return to safety

→ Approach-avoidance conflict

The agent moves via discrete actions on a 2D grid (up, down, left, right, stay).

Decisions are made sequentially based on current observations and learned policy.

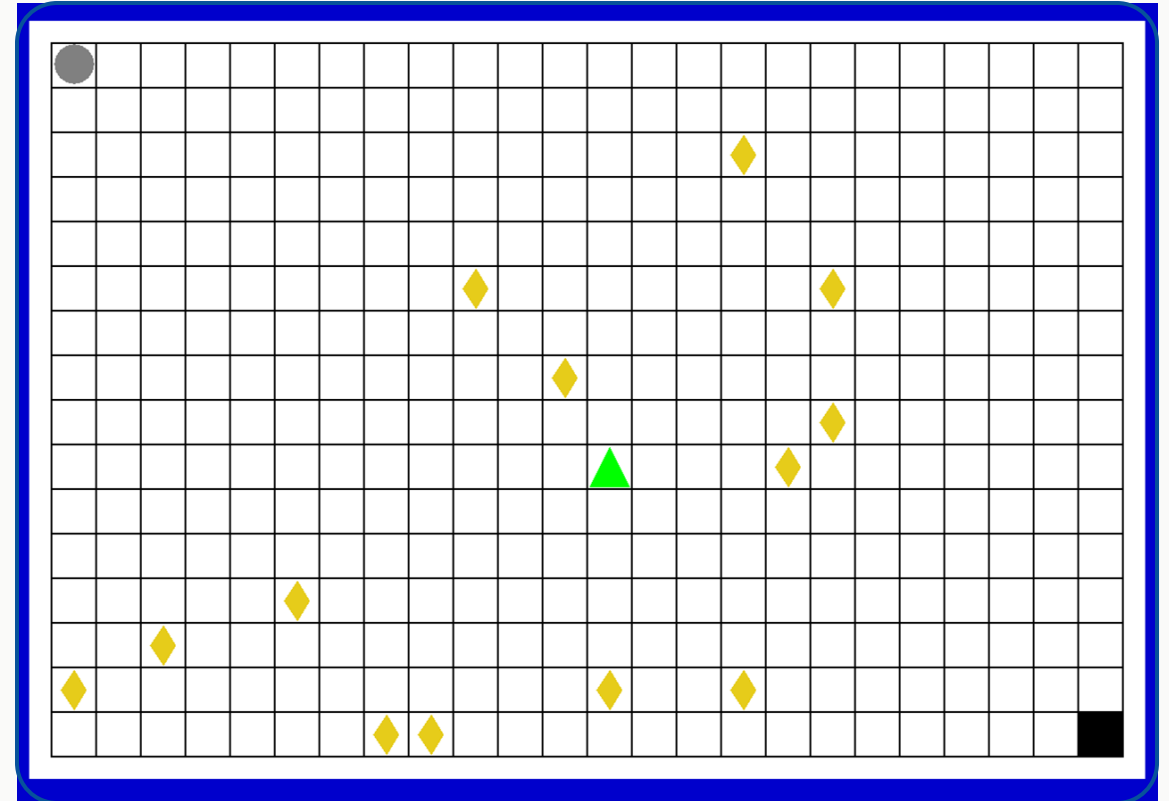


How Can We Model Constraints?

Unlike an optimal pathfinding algorithm, the agent has...

- limited perception,
- noisy observations,
- imperfect movement,
- and uncertainty about distant rewards.

→ Modeling human limitations



The objective is not optimality, but realistic and interpretable behavior.

03 Proposed Solutions

What Kinds of RL Can Be Used?

Based on the general outline of the environment and the agent design, we will implement

- Model-Free RL
 - “Learning directly from experience”
 - Pure trial and error
- Model-Based RL
 - “Learning using an internal model of the environment”
 - Explicit estimation of state transitions, future rewards and long-term consequences

Furthermore, we want to attempt implementing advanced approaches such as

- Representation-Based Models
- Deep RL

On that note, ideas from the optional papers may prove useful.

Model-Free RL

Q-Learning: How good is action A in state S?

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

- Off-policy: Learn best possible action independently of current exploration behavior
- Learns optimal policy over time
- Optimistic / risk-neutral

SARSA: Instead of assuming the best future action, learn from the action actually taken.

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma Q(s', a') - Q(s, a)]$$

- On-policy: Incorporate exploration behavior into learning
- Often produces safer or more cautious policies

Model-Based RL

Value iteration: Learning using an internal model of the environment

$$V(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V(s')]$$

- Uses knowledge of environment dynamics, plans ahead before acting
- More computationally expensive

Model-Free	Model-Based
Learns from rewards only	Learns environment dynamics
Faster/simpler	More computationally demanding
Habit-like behavior	Planning-like behavior

04 Evaluation

Evaluation: Comparing RL Metrics to Human Benchmarks

Standard RL Performance

Track total reward per episode, survival rate, and tokens collected per unit time.

Behavioral Analysis

Measure spatial behavior, including distance from the predator, time spent in the safe quadrant, and movement speed.

Bridging Models and Humans

Analyze greed versus safety (tokens collected before returning), catastrophic loss frequency, and proximity to the predator under high risk.

Human Benchmarking

Lab partners will play the foraging game to establish baseline human KPIs for direct comparison.

Interpretation

Understand how well model mechanisms can represent the assumed underlying neural characteristics

05 Work Plan

Work Plan

April 16 – May 21



Preparation

Read assigned papers.
Read about RL models.
Prepare proposal presentation.

June 15



Preliminary Demo

Implementation demo &
feedback session.

May 21 – July 23



Implementation & Analysis

Implement environment.
Implement models.
Analyze results.
Prepare final presentation.

July 31



Final Submission

Write report.
Clean up code.
Submit report and code.

May 21



Proposal Presentation

Deliver proposal presentation.

July 23



Final Presentation

Deliver final presentation.

06 References

References

(1) Image of the Hippocampus

Visual reference for neuroanatomical structures relevant to spatial memory.

[View Resource](#)

(2) MA-INF 4241: Threat Levels

Slides from the first meeting covering core definitions and experimental design.

(3) IBM: Reinforcement Learning

Introduction to basic RL concepts, agents, environments, and reward functions.

[View Resource](#)

(4) MA-INF 4241: Instructions

Guide and participation rules for the experiment.

Thank You!

Questions or feedback?

